# Experimental Contact III: The Network Game and the Gravitational Channel

The Real-versus-Complex Discrimination and the BMV Prediction in Finite Ring Cosmology

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**Abstract**

This note executes the two Tier-2 experimental contact points unblocked by the composite gate. *The network game (T4):* the entanglement-swapping protocol of the real-versus-complex discrimination — two independent sources emitting $|\Phi^+\rangle$, Alice measuring $\{\sigma_X, \sigma_Y, \sigma_Z\}$, Bob performing the Bell-state measurement, Charlie measuring the six $(\sigma_i \pm \sigma_j)/\sqrt{2}$ — is realised natively in FRC composites: the sources are synchronised drive-orbit pairs, and Bob's Bell-state measurement is the readout of his pair's *conserved relative offset and sign*, since the four orbit-sector states are exactly the Bell basis. In exact $\mathbb{Q}(\zeta_8)$ arithmetic, the full conditional correlation table of the FRC network equals the complex-quantum table — every entry, all $3 \times 6 \times 4$ setting–outcome combinations — so every linear score functional takes its complex-quantum value; the canonical witness evaluates to $6\sqrt{2} = 8.4853$, the network Tsirelson ceiling, strictly above the protocol's real-quantum bound 7.66. FRC thereby lands on the complex side of the experimental discrimination for the structural reason established in the companion: its amplitude ring is forced to $\mathbb{Z}[i]$ by the quarter-turn core. *The gravitational channel (T5):* in FRC gravity is phase synchronisation, so the gravitational interaction of two path-superposed clusters is an offset-diagonal conditional phase — coupled to the same conserved quantity that carries Bell correlations. The mechanism is verified exactly: zero coupling preserves product form; coupling phase $\varphi$ produces concurrence$^2 = \sin^2(\varphi/2)$ identically; the CHSH witness exceeds 2 for every $\varphi \neq 0$ and reaches $2\sqrt{2}$ at $\varphi = \pi$; the channel commutes with the drive, cannot signal, and obeys the which-path complementarity $V^2 + C^2 = 1$ as a ring identity. The forward prediction follows: BMV-class experiments will observe entanglement at exactly the Newtonian rate, with corrections bounded by the horizon and unobservable; a null result at that rate falsifies the framework's gravitational sector outright. All claims are machine-verified (`renou.py`, `bmv.py`; fourteen checks, all passing).

## 1 T4: the network game

### 1.1 The protocol and its FRC realisation

The discrimination experiments [4, 5, 6] use a three-party line network: independent sources $S_{AB}$, $S_{BC}$ each emit $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$; Alice measures one of $\{\sigma_X, \sigma_Y, \sigma_Z\}$; Bob performs the Bell-state measurement on his two qubits; Charlie measures one of the six $(\sigma_i \pm \sigma_j)/\sqrt{2}$, $i < j \in \{X, Y, Z\}$ [6]. Real-amplitude quantum theory with the standard composition rules caps the protocol's score at 7.6605; complex quantum theory reaches $6\sqrt{2} \approx 8.4853$, and the experiments decide for the complex side [4, 5, 6].

The FRC realisation assigns to every element of this protocol a native construction from the composite gate [2]: each source is a synchronised drive-orbit pair conditioned to winding doublets — the orbit state *is* $|\Phi^+\rangle$; Alice's and Charlie's analysers are forced readouts composed with relational repositionings; and Bob's Bell-state measurement receives the most natural reading in the framework:

**Proposition 1.1** (The BSM is the offset readout)**.** *The four Bell states of Bob's two doublet qubits are exactly the four synchronised orbit-sector states of his pair — the two values of the conserved relative offset, each with two signs.* Verified: the four orbit-sector states are mutually orthogonal with norm 2, forming the Bell basis exactly (`renou.py`, R1). *Bob's measurement is therefore the readout of a constant of motion: the entanglement-swapping operation, mysterious as a "projection onto entangled states," is in FRC the measurement of the one joint quantity the drive conserves.*

### 1.2 The result: table equality and the canonical score

**Theorem 1.2** (FRC reproduces the complex-quantum network exactly)**.** *In exact* $\mathbb{Q}(\zeta_8)$ *arithmetic* ($\sqrt{2} = \zeta_8 + \zeta_8^{-1}$, $\mathrm{i} = \zeta_8^2$)*: (i) conditioned on each of Bob's four outcomes b, the Alice–Charlie state is* $(\mathbb{1} \otimes \sigma_b)|\Phi^+\rangle$ *exactly — entanglement swapping; (ii)* $P(b) = \frac{1}{4}$ *for all b; (iii) the full conditional correlation table* $\langle A_x C_z \,|\, b\rangle$ *of the FRC network equals the complex-quantum table exactly, for all* $3 \times 6 \times 4$ *combinations. Consequently* every *linear score functional on the behaviour — the canonical witness included — takes its complex-quantum value: the witness evaluates to*

$$\mathcal{S} \;=\; 6\sqrt{2} \quad \textit{(ring identity: score}_{\mathrm{raw}} = 24(\zeta_8 - \zeta_8^3)\textit{)}, \tag{1}$$

*the network ceiling of three conditional CHSH expressions at* $2\sqrt{2}$ *per Bell outcome, strictly above the real-quantum bound* 7.6605. All statements machine-verified (`renou.py`, R2–R4).

The table-equality formulation is deliberately stronger than a single score: it certifies that FRC's predictions for this network are indistinguishable from complex quantum mechanics functional by functional, so the experimental verdicts [4, 5, 6] — violations of the real bound by tens of standard deviations — are verdicts *for* FRC. The structural reason was established in the companion [1]: the generic shared core of symmetry-complete shells is the quarter-turn group, the ledger is $\mathbb{Z}[\mathrm{i}]$, and a real-amplitude FRC would require shells without internal quarter-turns. The experiments that falsified real quantum theory falsified, in FRC vocabulary, the hypothesis that observers' frames lack the quarter-turn — which symmetry completeness forbids.

**Remark 1.3** (The independence assumption, localised)**.** The dissent in the real-versus-complex dispute [7] targets the tensor-independence axioms for the two sources. FRC localises this assumption as carrier arithmetic: the two orbit offsets are separately conserved, and the protocol's sources are independently prepared offset sectors (*verified, R5*); residual dependence between sources is measured by what their cycles share beyond the universal core — a computable quantity, not a postulate. The framework therefore does not merely pass the network test; it supplies a quantitative language for the very assumption under dispute.

## 2 T5: the gravitational channel

### 2.1 The mechanism

In FRC, gravitation is phase synchronisation on the substrate [3]: masses are winding rates, and the interaction of two systems is the mutual adjustment of their phases. For two locked clusters each placed in a two-path superposition — the BMV configuration [8, 9] — the gravitational interaction acts as a branch-pair-dependent phase, accumulating at the Newtonian rate $\varphi(t) = \frac{Gm_1m_2}{\hbar}\left(\frac{1}{d} - \frac{1}{d'}\right) t$ with the potential supplied by the gravity companion's derived lattice Green's function. The decisive structural fact: this coupling is *offset-diagonal* — it acts on the same conserved joint quantity whose readout is the Bell-state measurement of Section 1. Gravity, in this framework, talks directly to the degree of freedom that carries entanglement.

**Proposition 2.1** (The channel entangles, exactly)**.** *For the joint branch state of two clusters with coupling phase $\varphi = 2\pi k/80$: (i) at $\varphi = 0$ the state remains an exact product; (ii) for $\varphi \neq 0$ the concurrence obeys $C^2 = \sin^2(\varphi/2)$ as a cyclotomic ring identity, over the full sweep; (iii) the CHSH witness reaches $2\sqrt{1+\sin^2(\varphi/2)} > 2$ for every $\varphi \neq 0$, attaining the Tsirelson value $2\sqrt{2}$ at $\varphi = \pi$ — consistency with the composite gate; (iv) the coupling commutes with the drive, Alice's marginal is invariant under arbitrary local operations on Bob's side (no signalling), and the interferometric visibility obeys the complementarity*

$$V^2 + C^2 = 1 \tag{2}$$

*exactly — the which-path trade-off that constitutes the BMV observable pair.* All machine-verified (`bmv.py`, B1–B4).

### 2.2 The forward prediction

**Remark 2.2** (FRC prediction for BMV-class experiments)**.** The framework predicts, with no adjustable parameter: *gravitationally induced entanglement forms, at exactly the Newtonian rate*, with visibility and concurrence trading off as $V^2 + C^2 = 1$; corrections to the rate are bounded by the coherence horizon ($\sim 1/\sqrt{\Omega}$) and are unobservable at any proposed mass and separation. This places FRC on the opposite side from semiclassical models, which predict no entanglement [10], and makes the prediction falsifiable in both directions: observation of entanglement at a non-Newtonian rate, or a robust null result at the Newtonian rate, each falsifies the framework's gravitational sector outright — there is nothing to retune. With constrained-dynamics variants relaxing the experimental requirements [11], this is the programme's primary dated forward prediction, filed before the first data.

## 3 Status

**Closed.** T4: the network protocol realised natively, the full conditional table equal to the complex-quantum table in exact $\mathbb{Q}(\zeta_8)$ arithmetic, the canonical witness at the ceiling $6\sqrt{2}$ above the real bound; the BSM identified as the offset readout. T5: the gravitational channel's entangling mechanism verified exactly, the complementarity identity established, the forward prediction stated. Fourteen machine checks across the two suites, all passing.

**Residue, named.** Alice's and Charlie's equatorial analysers and the polar readout are corpus-native today (the composite gate's repositionings, quotient readouts, and absorption readouts); the mixed-polar analyser families — bases at 45° between the winding axis and the equator — are ledger-native (legitimate orthogonal measurement vectors) but their explicit construction from corpus operations (metaplectic compositions on the host cycle) is an open construction item. It affects which Subjects can *implement* the protocol, not what the protocol predicts.

**The tier list after this note.** T1–T5 closed. Remaining: T6 (the parameter-free decoherence-floor contour against the matter-wave interferometry roadmap), T7 (the locked/unlocked coupling phenomenology against equivalence-principle bounds), T8 (the amplitude-granularity ceiling), T9 (qudit emulation), and the standing $\Omega$-ledger consolidation.

## Reproducibility

The scripts `renou.py` and `bmv.py` reside in the `validation/` folder of the companion draft, slated for https://github.com/gamayos/frc-quantum-numerics. Recorded output: fourteen `PASS` lines and the terminal lines `renou: all checks passed`, `bmv: all checks passed`.

## References

[1] Y. Akhtman. *Quantum Observation in Finite Ring Cosmology.* Companion draft, June 2026.

[2] Y. Akhtman. *Experimental Contact II: The Composite Gate.* Companion note, June 2026.

[3] Y. Akhtman and E. Voether. *Gravitation as Phase Synchronisation on a Finite Substrate.* Preprint, 2026.

[4] M.-O. Renou et al. Quantum theory based on real numbers can be experimentally falsified. *Nature*, 600:625–629, 2021.

[5] M.-C. Chen et al. Ruling out real-valued standard formalism of quantum theory. *Physical Review Letters*, 128:040403, 2022.

[6] Z.-D. Li et al. Testing real quantum theory in an optical quantum network. *Physical Review Letters*, 128:040402, 2022.

[7] T. Hoffreumon and M. P. Woods. Quantum theory based on real numbers cannot be experimentally falsified. arXiv:2603.19208, 2026.

[8] S. Bose et al. Spin entanglement witness for quantum gravity. *Physical Review Letters*, 119:240401, 2017.

[9] C. Marletto and V. Vedral. Gravitationally induced entanglement between two massive particles is sufficient evidence of quantum effects in gravity. *Physical Review Letters*, 119:240402, 2017.

[10] Absence of gravitationally induced entanglement in certain semi-classical theories of gravity. arXiv:2510.20991, 2025.

[11] Gravity-induced entanglement under constrained dynamics. arXiv:2605.00967, 2026.